



Lesson 4.7 — Adding and Subtracting Rational Expressions

Big idea: To add or subtract fractions, you need a common denominator — same idea here. Find the LCD, rewrite each fraction over it, combine the numerators.

Quick review

Step 1: factor every denominator.

Step 2: the LCD is each unique factor raised to the highest power that appears.

Step 3: rewrite each fraction so they all have the LCD.

Step 4: add or subtract numerators (carefully — distribute the negative sign for subtraction).

Step 5: simplify the resulting fraction (factor and cancel any common factor).

Worked example

Worked example. Compute $2/(x - 1) - 3/(x + 2)$.

$$\text{LCD} = (x - 1)(x + 2). [2(x + 2) - 3(x - 1)] / [(x - 1)(x + 2)] = (2x + 4 - 3x + 3) / [...] = (-x + 7) / [(x - 1)(x + 2)].$$

Warm-ups

1. Compute $3/x + 5/x$.
2. Compute $1/2 + 1/3$.
3. Compute $1/x - 1/(x + 1)$.

Practice

1. Compute $(2x)/(x - 3) + 4/(x - 3)$.
2. Compute $3/(x + 1) - 2/(x - 4)$.
3. Compute $(x)/(x^2 - 1) + 1/(x - 1)$.

Challenge

1. Simplify $[(x + 1)/(x - 2)] - [(x - 3)/(x + 2)]$.
2. Add $(1)/(x^2 - 4) + (2)/(x^2 - 4x + 4)$ and simplify.



Answer key — Lesson 4.7

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $8/x$.
2. $5/6$.
3. $(x + 1 - x)/[x(x + 1)] = 1/[x(x + 1)]$.

Practice

1. $(2x + 4)/(x - 3)$.
2. $[3(x - 4) - 2(x + 1)]/[(x + 1)(x - 4)] = (x - 14)/[(x + 1)(x - 4)]$.
3. $x^2 - 1 = (x - 1)(x + 1)$. LCD: $(x - 1)(x + 1)$. $[x + (x + 1)]/[(x - 1)(x + 1)] = (2x + 1)/[(x - 1)(x + 1)]$.

Challenge

1. LCD = $(x - 2)(x + 2)$. $[(x + 1)(x + 2) - (x - 3)(x - 2)]/[(x - 2)(x + 2)] = [(x^2 + 3x + 2) - (x^2 - 5x + 6)]/[...] = (8x - 4)/[(x - 2)(x + 2)] = 4(2x - 1)/[(x - 2)(x + 2)]$.
2. Factor: $1/[(x - 2)(x + 2)] + 2/(x - 2)^2$. LCD: $(x - 2)^2(x + 2)$. $[(x - 2) + 2(x + 2)] / [(x - 2)^2(x + 2)] = (3x + 2)/[(x - 2)^2(x + 2)]$.